Majid Dousti

Product Engineer | Mechanical Design, Validation & Production Launch
Dayton, OH (open to relocation) — Permanent Resident — (434) 326-2187 — Dousti.explore@gmail.com
linkedin.com/in/majid-dousti — Google Scholar

Product engineer with 5+ years taking mechanical products from CAD design and prototyping through validation and production launch in automotive glass, industrial water treatment, and furniture manufacturing. Currently supports Ford windshield, sidelite, and roof programs through design and production verification, special characteristics, and control plans. Designs in SolidWorks, Autodesk Inventor, and CATIA, and combines GD&T, tolerance analysis, and root cause analysis with data analysis in Python and SQL. M.S. Mechanical Engineering; M.S. Computer Science expected Fall 2027.

Experience

System and Quality Engineer May 2026 – Current
Fuyao Glass America Dayton, OH

- **Design verification:** Tracked Ford windshield DV to 103 of 110 DV tests (94%) passed on schedule; 7 long-duration tests aligned with Ford, 2 of them delayed by discontinued material.
- **Production verification:** Prepared the PV test list following the DVP&R, cross-checked against Ford specifications, regulations, and standards.
- **Special characteristics:** With the team, finalized the second version of the SCCAF and Control Plan for windshield, sidelite, and roof, signed by Ford and Fuyao for the new production line.
- **Edge stress defect:** Resolved a 4-month windshield edge stress defect in 20 days through root cause studies and three debug trials with furnace, engineering, and quality; locked the resulting parameters as the production standard.
- **Optical investigation:** Led a camera-zone transmissivity investigation with lab, quality, and engineering teams; the technical report resolved core misunderstandings with Ford's camera team.
- **Measurement:** Standardized checking-fixture measurement exports into reports against specification limits using AI-assisted tools; supported gate reviews with measurement system self-assessments.
- **Checking fixtures:** Calibrated and updated windshield and sidelite checking fixtures for the production line.
- **Safe launch:** Built a live yield and defect dashboard for a 90-day Rivian safe launch as yield rose from 64% to 85% in two months; rolled it out to other programs, where it remains in use.
- **New production line:** Worked with production and tooling teams on line integration, targets, milestones, bottlenecks, and facility readiness.
- **Customer support:** Technical contact for Ford and Sales on deliverables, product changes, and specifications within contract scope.
- **Recognition:** Member of the team recognized as a Ford Q1 Preferred Quality Supplier.

Teaching Assistant, Computer Programming Jan 2025 – May 2026
Wright State University Dayton, OH

- **Teaching:** Taught Computer Programming to about 45 students.
- **Labs:** Ran the course lab sessions.
- **Office hours:** Held office hours for students.

Product Development Engineer Nov 2023 – Jan 2025
AmeriWater Co. Dayton, OH

R&D, testing, and validation

- Led two models in the AWRO reverse osmosis family from material selection and prototyping through production release; finished and tested both prototypes 15 days ahead of schedule.
- Led full-system assembly and commissioning of the AWRO 1200 prototype, including plumbing layout, BOM sourcing, and final validation.
- Validated thermal performance through PAA testing on RO models (1200 to 19200), capturing 50+ samples per unit for FDA-regulated compliance.
- Diagnosed and corrected system failures, including pump retry and thermal trip, during firmware testing and verification.
- Integrated control systems with electrical and PLC engineers and validated them against customer requirements.

Design and floor work

- Led design revisions for carbon filter assemblies and tank systems in Autodesk Inventor, keeping them compatible with existing BOMs and assembly.
- Applied GD&T tolerance checks to production drawings for pressure vessels, pumps, and plumbing layouts.
- Resolved downtime with manufacturing and maintenance teams by troubleshooting leaks, tubing, GFCI errors, and defective pumps.

Documentation and suppliers

- Created part numbers, BOMs, and drawings in Intuitive ERP; updated operation manuals and test methods.

- Evaluated materials and sourced custom parts with external vendors.

Associate Technical Program Coordinator Oct 2019 – Dec 2021
Meranti Co. Tehran, Iran

- Led the product engineers added as the team grew, while continuing product design work.
- Led the team that built BOMs for all company products and the company's CRM system.
- Delivered 3 projects with design and first prototype on schedule.
- Sourced parts with multiple vendors and ran incoming quality control with QA; gathered customer requirements and dispatched installation teams.

Product Engineer April 2018 – Oct 2019
Meranti Co. Tehran, Iran

- Joined at the company's start and owned its product designs and full product catalogs for office furniture.
- Created 100+ fabrication and assembly drawings in SolidWorks and AutoCAD, translating concepts into manufacturable parts.
- Performed mechanical design and analysis for modular furniture assemblies, ensuring structural integrity and manufacturability.
- Programmed and operated CNC machines in Mastercam for 6 months, creating toolpaths for furniture panels in high-volume production.
- Developed machining programs that optimized cutting parameters, tool selection, and cycle times.
- Applied GD&T and tolerance analysis to resolve fit issues between interfacing components.
- Trained installation teams and supported products from concept layout through installation validation and after-sales.

Technical Program Coordinator April 2016 – March 2018
Samsung / Regional Office Tehran, Iran

- Supported an Android and iOS app that reached 65,000+ registrations in launch week and 80,000+ users across about 15,000 branches.
- Managed weekly meetings with the app teams using real market data, and brought market feedback to design teams.

Research and Certifications

Digital twin of synchronized laser arrays (FEM, MOOSE), Advanced Manufacturing Technology 2022
Neural surrogate for melt pool modeling (3D CNN), M.S. Mechanical Engineering thesis 2025
OASiS Rapid Certification in semiconductor manufacturing, Intel-endorsed 2025

Education

Master of Computer Science, Wright State University, GPA: 3.8/4 Expected Fall 2027
M.S., Mechanical Engineering, Wright State University, GPA: 3.91/4 July 2025
Bachelor's Degree, Civil Engineering, University of Zanjan 2008 – 2013

Skills

CAD: SolidWorks (3+ years), Autodesk Inventor, CATIA, AutoCAD (2D/3D), sheet metal design
Design & standards: GD&T, tolerance analysis, engineering drawings, technical specifications, engineering change orders, design of experiments (DOE), DFM/DFA
Validation & quality: DV and PV, DVP&R, SCCAF, control plans, measurement system assessment, safe launch, root cause and failure analysis
Manufacturing: new line integration, CNC programming (Mastercam), fabrication drawings, fixture concepts, shop-floor troubleshooting
Analysis & software: Python, MATLAB, C++, LabVIEW, SQL, Tableau, Excel dashboards, FEM and thermal simulation (MOOSE)
Systems: BOMs, ERP (Intuitive), system integration, validation protocols, MS Project, Smartsheet

Selected Honors

Reviewer, Materials Science and Engineering: A (Elsevier), May 2022 – Present | Travel Award, Material Research Symposium, Seattle, WA, April 2025 | Travel Award, 10th International Conference on Multiscale Materials Modeling, Baltimore, MD, Oct 2022 | SolidWorks Instructor, Intel Summer Internship Program for Minorities, Wright State University, July 2023